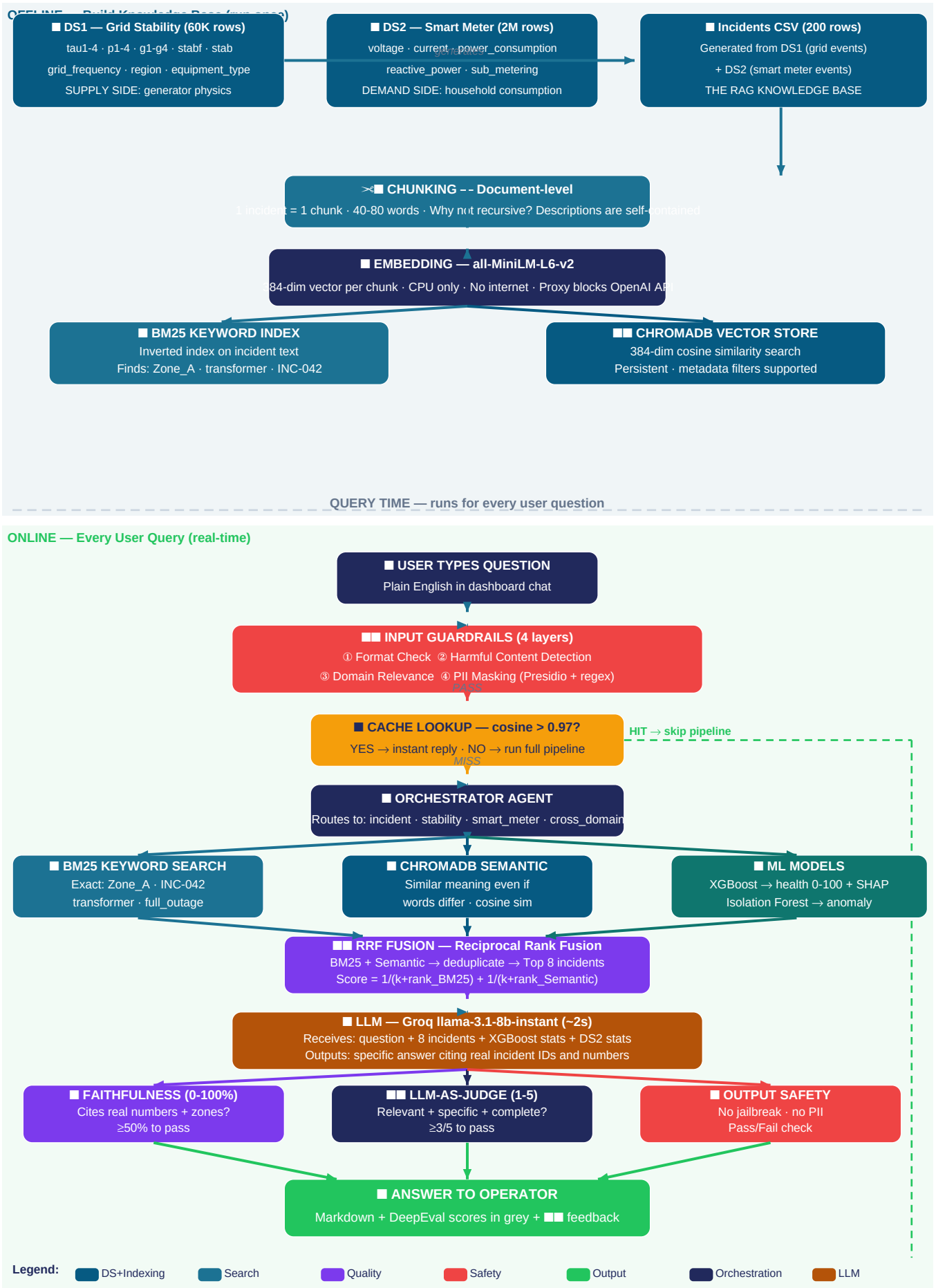


SGEIA — RAG Pipeline Architecture

Complete flow: CSV Load → Chunking → Embedding → Indexing → ChromaDB → Hybrid Search → LLM Processing → DeepEval → Output



STEP-BY-STEP REFERENCE — What Happens at Each Stage

Step	Stage	What Happens — Technical Detail
1	■ Load CSV Files	pandas read_csv() loads all 3 datasets at startup. DS1 (60K rows) and DS2 (2M rows) are read with only required columns. Incident CSV is loaded into in-memory cache so dashboard never waits for disk.
2	>■ Chunking	Technique: Document-level (1 incident = 1 chunk, 40-80 words). Why NOT recursive/sliding window chunking? Incident descriptions are already self-contained operational narratives. Splitting would break the context of each event.
3	■ Embedding	Model: all-MiniLM-L6-v2 (sentence-transformers). Converts each chunk to a 384-dimensional vector. Runs on CPU, no internet needed. Why NOT OpenAI text-embedding-3? Corporate McAfee proxy blocks the API call.
4	■ BM25 Indexing	Library: rank-bm25. Tokenises each incident description, builds inverted keyword index. Fast exact matching for: Zone_A, Zone_B, Zone_C, Zone_D, INC-042, transformer, full_outage, critical.
5	■■ ChromaDB Storage	All 384-dim vectors + metadata saved to disk (chroma_db/ folder). Persists across restarts. Supports simultaneous metadata filters: WHERE region=Zone_A AND severity=critical AND equipment_type=transformer.
6	■■■ Input Guardrails	4 layers: ① Format check (5-1000 chars) ② Harmful content regex (violence/self-harm/weapons/illegal) ③ Domain keyword allowlist (50+ grid terms) ④ PII masking via Microsoft Presidio + regex fallback.
7	■ Cache Lookup	ChromaDB query_cache collection. If new query has cosine similarity > 0.97 with a cached query → return instantly. Operator ■ ratings and high-scoring answers are also cached.
8	■ Orchestrator Routing	LLM Query Router (GPT-4o-mini or llama) classifies: incident/stability/smart_meter/cross_domain/out_of_scope. Detects metadata filters from query text. Routes to only the needed agents.
9	■ BM25 Search	Searches tokenised incident text for keyword matches. Returns ranked list. Best for: exact zone names, incident IDs, equipment types, outage event names.
10	■ ChromaDB Semantic	Embeds the user query → 384-dim vector → cosine similarity search. Returns semantically similar incidents even when words differ. Applied metadata filters from step 8.
11	■■ RRF Fusion	Score = $1/(k+\text{rank_BM25}) + 1/(k+\text{rank_semantic})$, k=60. Merges both lists, removes duplicates, selects Top 8. Why hybrid? +15-20% recall improvement over single method.
12	■ ML Analysis	XGBoost on DS1: classifies stable/unstable, outputs health_score=(1-P)×100, SHAP values show which feature caused it. Isolation Forest on DS2: detects anomalous consumption patterns (unsupervised).
13	■ LLM Processing	Groq llama-3.1-8b-instant receives: question + 8 incidents (with IDs/descriptions) + XGBoost health score + SHAP values + DS2 stats + widget context. Outputs grounded answer citing real numbers.
14	■ DeepEval — 3 Scores	① Faithfulness (0-100%): cites real numbers/zones/IDs? ② LLM-as-Judge (1-5): relevant + specific + complete? ③ Output Safety: no jailbreak, no PII, no prompt leak. Shown in grey below answer.
15	■ Output Guardrails	Final scan: jailbreak patterns, system prompt fragments (=== markers), PII in output. Answer blocked if harmful content found. Short answers replaced with fallback message.
16	■ Cache Store	If faithfulness ≥ 50% or operator gives ■ → save to ChromaDB query_cache. Future identical questions answered in <100ms without running the pipeline.
17	■ Answer to Operator	Displayed in chat bubble with: markdown formatting (bold, bullets), DeepEval scores in grey (Faithfulness % · LLM Judge /5 · Safety ■), ■/■ feedback buttons.

Key Numbers: DS1: 60,000 rows · DS2: 2,000,000 rows · Incidents: 200 · Embeddings: 384-dim · Top-K: 8 results · Cache threshold: cosine>0.97 · Judge pass: ≥3/5 · Faithfulness pass: ≥50% · LLM response: ~2s · Dashboard refresh: 60s